Tugas Sistem Operasi

“Redhat Academy Course 8”

Disusun Oleh:

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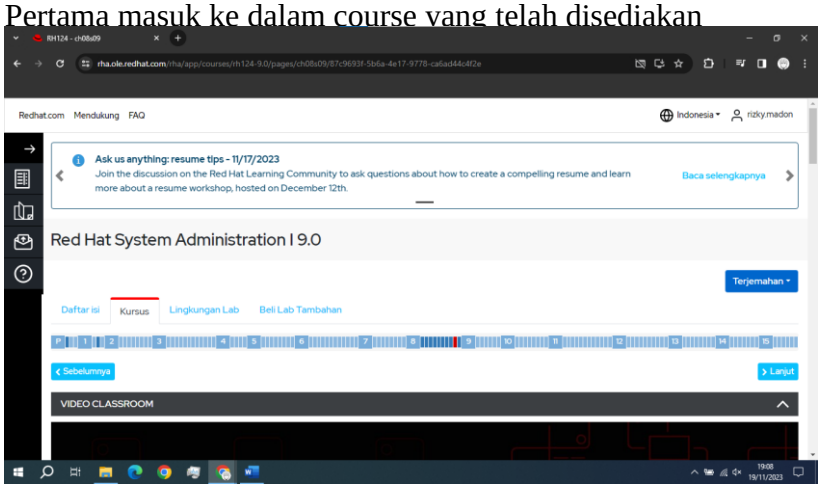
Dosen Pengampu :

Ahmad Heryanto, S.Kom, M.T.

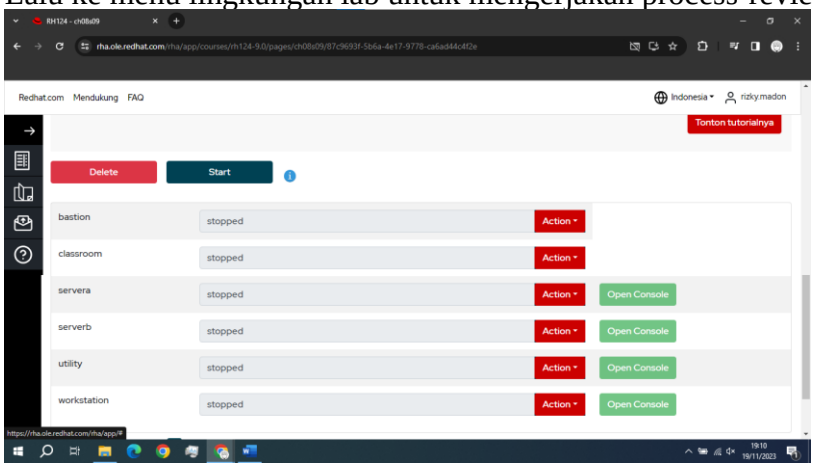
JURUSAN SISTEM KOMPUTER
FAKULTAS ILMU KOMPUTER
UNIVERSITAS SRIWIJAYA

TAHUN AJARAN 2023/2024

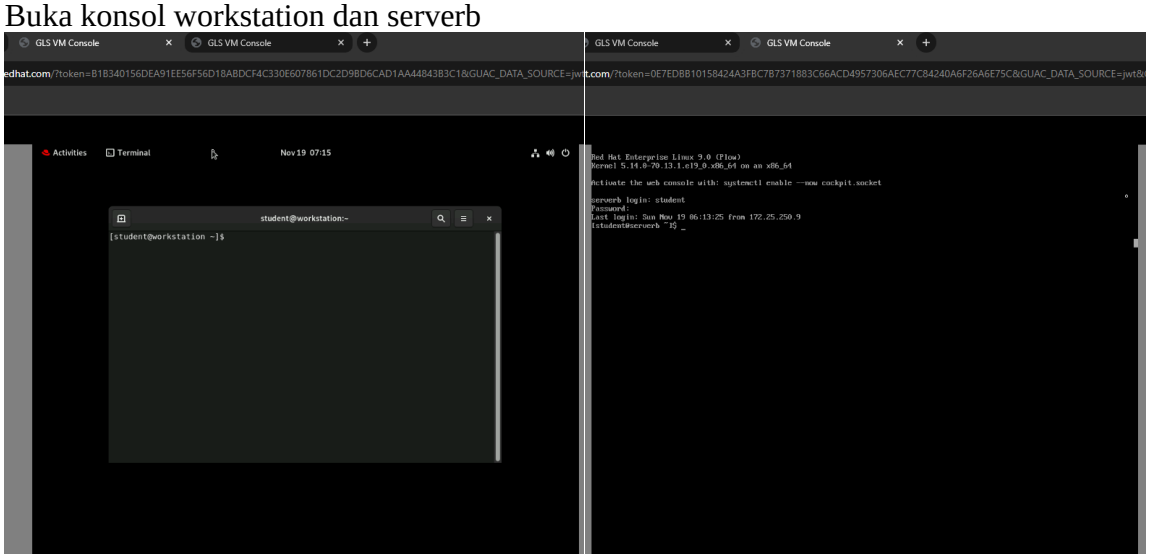
1. Pertama masuk ke dalam course yang telah disediakan



2. Lalu ke menu lingkungan lab untuk mengerjakan process-review



3. Buka konsol workstation dan serverb



4. Mulai Latihan di workstation dengan command berikut

```
[student@workstation ~]$ lab start processes-review

Starting lab.

 · Checking lab systems ..................................... SUCCESS

[student@workstation ~]$
```

5. Sambungkan ssh serverb ke workstation

```
[student@workstation ~]$ ssh student@serverb
Warning: Permanently added 'serverb' (ED25519) to the list of known hosts.
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Sun Nov 19 05:58:36 2023
[student@serverb ~]$
```

6. Buat direktori bin lalu buat process101 menggunakan vim, gunakan chmod untuk mengubah izinnya

```
[student@serverb ~]$ mkdir /home/student/bin
[student@serverb ~]$ vim /home/student/bin/process101
[student@serverb ~]$ chmod +x /home/student/bin/process101
```

7. Ubah isi dari process101 menggunakan vim

```
#!/bin/bash
while true; do
        var=1
        while [[ var -lt 50000 ]]; do
                var=$(($var+1))
        done
        sleep 1
done
~
```

8. Gunakan command top untuk melihat proses yang ada

```
top - 06:10:20 up 27 min,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 116 total,   1 running, 115 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   1774.8 total,   1445.4 free,    183.0 used,    146.4 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   1442.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  171404  15680  10324 S   0.0   0.9   0:01.09 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root      20   0       0      0      0 I   0.0   0.0   0:00.04 kworker+
      9 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_perc+
     10 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_tas+
     11 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_tas+
     12 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_tas+
     13 root      20   0       0      0      0 S   0.0   0.0   0:00.00 ksoftir+
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.01 rcu_pre+
```

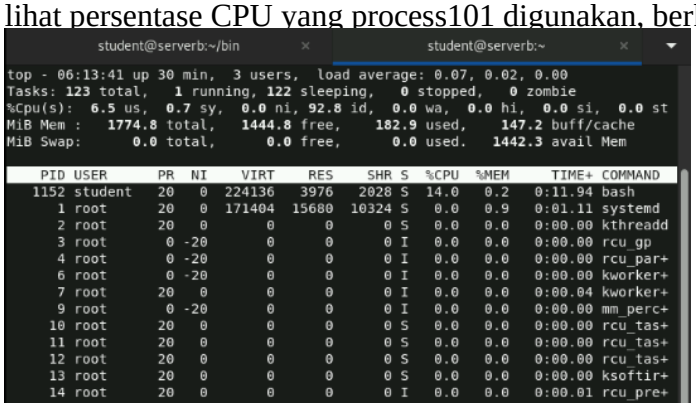
9. Untuk mengecek jumlah cpu gunakan command ini

```
[student@serverb ~]$ grep "model name" /proc/cpuinfo | wc -l
2
```

10. Jalankan proses101 dengan command berikut

```
[student@serverb bin]$ process101 &
[1] 1152
```

11. Membuat terminal Baru pada linux untuk melihat sistem yang berjalan pada linux dan lihat persentase CPU yang process101 digunakan, berkisar antara 10% hingga 15%



12. Untuk melihat persentase memori digunakan, tekan tombol m

```
top - 06:14:24 up 31 min,  3 users,  load average: 0.08, 0.03, 0.01
Tasks: 123 total,   1 running, 122 sleeping,   0 stopped,   0 zombie
%Cpu(s):  7.2 us,  1.0 sy,  0.0 ni, 91.7 id,  0.0 wa,  0.2 hi,  0.0 si,  0.0 st
MiB Mem : 18.8/1774.8   [|||||||||                                          ]
MiB Swap:  0.0/0.0      [                                                   ]

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
```

13. Untuk melihat persentase cpu digunakan, tekan tombol t

```
student@serverb:~/bin                    student@serverb:~
top - 06:16:02 up 32 min,  3 users,  load average: 0.01, 0.02, 0.00
Tasks: 122 total,   2 running, 120 sleeping,   0 stopped,   0 zombie
%Cpu(s):   5.5/0.5     6[|||                                                ]
MiB Mem : 18.8/1774.8   [|||||||||                                          ]
MiB Swap:  0.0/0.0      [                                                   ]

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
```

14. Untuk menyimpan konfigurasi, tekan shift+w, lalu cek dengan command berikut

```
[student@serverb ~]$ ls -l /home/student/.config/procps/toprc
-rw-r--r--. 1 student student 1146 Nov 19 06:16 /home/student/.config/procps/top
rc
```

15. Salin proses101 ke proses102

```
[student@serverb bin]$ cp process101 process102
```

16. Gunakan vim untuk merubah isi menjadi 100000

```
#!/bin/bash
while true; do
        var=1
        while [[ var -lt 100000 ]]; do
                var=$(($var+1))
        done
        sleep 1
done
```

17. Jalankan proses tsb, gunakann command jobs untuk melihat proses yang berjalan

```
[student@serverb bin]$ cp process101 process102
[student@serverb bin]$ vim process102
[student@serverb bin]$ process102 &
[2] 3579
[student@serverb bin]$ jobs
[1]-  Running                 process101 &
[2]+  Running                 process102 &
```

18. Kemudian Mengecek lagi pada top terminal satunya bahwa proses sedang berjalan dan menggunakan sebagian besar daya Cpu.

```
top - 06:34:29 up 51 min,  3 users,  load average: 0.85, 0.73, 0.38
Tasks: 129 total,   3 running, 126 sleeping,   0 stopped,   0 zombie
%Cpu(s):  49.2/4.5    54[|||||||||||||||||||||||||||||                      ]
MiB Mem : 19.6/1774.8   [||||||||||                                         ]
MiB Swap:  0.0/0.0      [                                                   ]
```

19. Copy lagi untuk membuat proses ketiga

```
[student@serverb bin]$ cp process101 process103
[student@serverb bin]$ vim process103
```

20. Edit dengan mengubah menjadi 800000

```
#!/bin/bash
while true; do
        var=1
        while [[ var -lt 800000 ]]; do
                var=$(($var+1))
        done
        sleep 1
done
```

21. Jalankan lalu lihat proses yang berjalan

```
[student@serverb bin]$ process103 &
[3] 4342
[student@serverb bin]$ jobs
[1]   Running                 process101 &
[2]-  Running                 process102 &
[3]+  Running                 process103 &
```

22. Kemudian masuk ke dalam rootuser lalu hentikan procces101 dan ketik ps jT untuk melihat status

```
[student@serverb bin]$ su
Password:
[root@serverb bin]# pkill -SIGSTOP process101
[root@serverb bin]# ps jT
   PPID     PID    PGID     SID TTY        TPGID STAT   UID   TIME COMMAND
   1107    1108    1108    1108 pts/0       4707 Ss    1000   0:00 -bash
   1108    1152    1152    1108 pts/0       4707 S     1000   3:20 -bash
   1108    3579    3579    1108 pts/0       4707 S     1000   0:49 -bash
   1108    4342    4342    1108 pts/0       4707 R     1000   0:42 -bash
   1108    4478    4478    1108 pts/0       4707 S        0   0:00 su
   4478    4498    4498    1108 pts/0       4707 S        0   0:00 bash
   3579    4701    3579    1108 pts/0       4707 S     1000   0:00 sleep 1
   1152    4703    1152    1108 pts/0       4707 S     1000   0:00 sleep 1
   4498    4707    4707    1108 pts/0       4707 R+       0   0:00 ps jT
```

23. Kill semua process101,102,103 lalu Exit

```
[root@serverb bin]# pkill process101
[root@serverb bin]# pkill process102
[root@serverb bin]# pkill process103
[root@serverb bin]# exit
```

24. Lihat penilaian apakah Semuanya PASS

```
[student@workstation ~]$ lab grade processes-review

Grading lab.

 · Checking lab systems ..................................................... SUCCESS
 · Verifying that process101 is not running ................................. SUCCESS
 · Verifying that process102 is not running ................................. SUCCESS
 · Verifying that process103 is not running ................................. SUCCESS

Overall lab grade: PASS
```

25. Tutup work station agar tidak mengganggu latihan selanjutnya

```
[student@workstation ~]$ lab finish processes-review

Finishing lab.

 · Checking lab systems ..................................................... SUCCESS
 · Terminating processes ................................................... SUCCESS
 · Deleting /home/student/bin .............................................. SUCCESS
 · Deleting /home/student/.config/procps/toprc ............................. SUCCESS
```